

Exploratory Data Analysis Report



Mental Health Text Classification Dataset

Preparing Data for Traditional ML, Deep Learning & Transformer-Based Models

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Dataset: train_split.csv | Samples: 34,601 | Classes: 4

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1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Mental Health Text Classification Dataset, which contains 34,601 text samples labeled across four categories: Normal (36.7%), Depression (29.3%), Suicidal (22.7%), and Anxiety (11.3%). The analysis is specifically designed to prepare the data for three distinct modeling approaches: Traditional Machine Learning (TF-IDF + Logistic Regression/SVM/Random Forest), Deep Learning (LSTM/GRU/CNN for text), and Transformer-based Fine-tuning (BERT). The dataset presents several critical challenges including class imbalance with a 3.26:1 ratio between the majority and minority classes, significant text length variability ranging from 8 to 38,785 characters, and substantial vocabulary overlap between mental health categories (especially Depression-Suicidal cosine similarity of 0.919).

Key findings reveal that mental health categories (Depression, Suicidal, Anxiety) exhibit considerably longer texts than the Normal class, with mean word counts of 107-126 versus 24 for Normal. The vocabulary overlap analysis shows that Depression and Suicidal classes share 91.9% TF-IDF centroid similarity, making them the most confusable pair. Data quality issues affect approximately 11.5% of samples with non-ASCII Unicode characters, 2.66% extremely short texts, and 1.79% containing URLs. These findings directly inform preprocessing strategies and model architecture decisions for each of the three modeling paradigms, with specific recommendations detailed in Sections 7 through 9.

2. Dataset Overview

2.1 Basic Information

The dataset consists of 34,601 text-label pairs stored in CSV format with two columns: 'text' containing the raw text content and 'label' containing the class designation. The dataset has no missing values and no exact duplicate rows, which is a positive indicator of data integrity. However, the presence of near-duplicate texts with different labels or minor variations cannot be ruled out without further fuzzy matching analysis. The four classes represent distinct mental health states: Normal (non-mental-health-related text), Depression, Suicidal ideation, and Anxiety, making this a multi-class classification problem with significant real-world implications for mental health screening applications.

Property	Value
Total Samples	34,601
Number of Classes	4
Missing Values (Text)	0
Missing Values (Label)	0
Exact Duplicate Rows	0
Duplicate Texts	0

Property	Value
Unique Words (raw)	40,593
Unique Words (after stopwords removal)	40,288
Mean Word Count	79.3
Median Word Count	48
Imbalance Ratio (max/min)	3.26

2.2 Class Distribution Analysis

The class distribution exhibits moderate imbalance, with the Normal class being the most represented at 36.7% (12,706 samples) and Anxiety being the least represented at 11.3% (3,901 samples). The imbalance ratio of 3.26:1 (Normal to Anxiety) is not extreme but is significant enough to impact model training, particularly for minority class recall. Depression accounts for 29.3% (10,154 samples) and Suicidal for 22.7% (7,840 samples). This distribution reflects a realistic scenario where normal social media text is more prevalent than mental health-related content, and within mental health categories, anxiety expressions may be shorter and less frequently shared in long-form posts compared to depression narratives.

The imbalance has direct implications for model training: without intervention, classifiers will tend to favor the Normal class at the expense of Anxiety. Techniques such as class-weighted loss functions, SMOTE oversampling for the minority class, or strategic under-sampling of the majority class should be considered. For BERT fine-tuning specifically, the class imbalance can be addressed through weighted cross-entropy loss or focal loss, while for traditional ML, SMOTE or class-weighted SVM are recommended approaches.

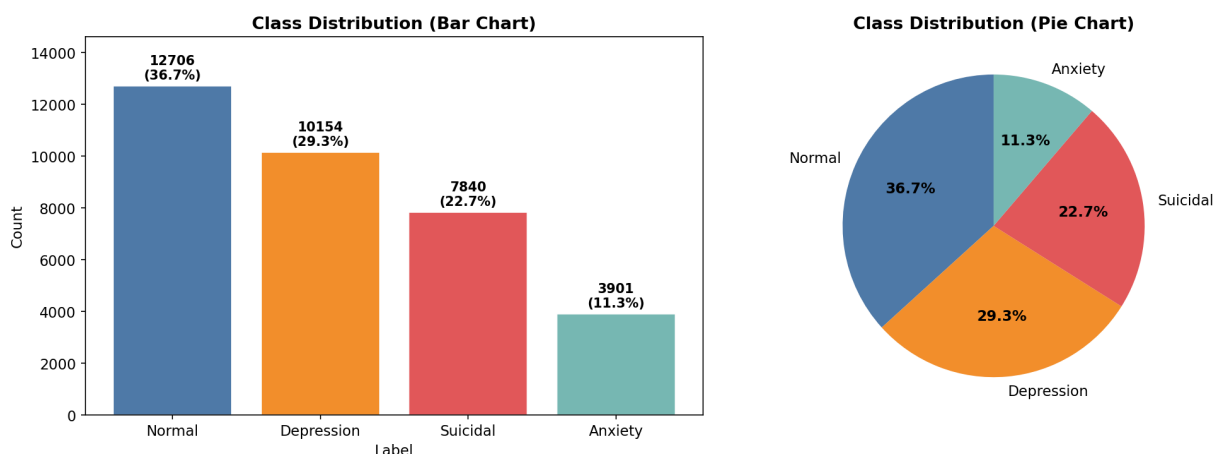


Figure 2-1: Class Distribution — Bar Chart and Pie Chart

3. Text Length & Statistical Analysis

3.1 Character and Word Count Distributions

Text length analysis reveals dramatic differences between the Normal class and mental health categories. The Normal class has a median character length of only 60 (mean 128), while Depression (median 434, mean 539), Suicidal (median 396, mean 551), and Anxiety (median 502, mean 665) are all significantly longer. This pattern is consistent with the nature of mental health expression: individuals sharing emotional distress tend to write longer, more detailed posts as a form of self-expression and help-seeking. The Normal class, by contrast, includes many short social media updates, casual remarks, and brief statements. The word count distribution follows a similar pattern, with Normal texts averaging 24 words versus 107-126 for mental health categories.

The distributions are all right-skewed with long tails, indicating the presence of very long texts across all classes. The maximum text length is 38,785 characters (6,353 words) in the Normal class, which is an extreme outlier. The standard deviations are large relative to the means (e.g., Depression has a standard deviation of 536 characters against a mean of 539), confirming high variability. For modeling purposes, this variability necessitates careful handling of sequence lengths: padding and truncation strategies for deep learning models, and feature engineering for traditional ML approaches.

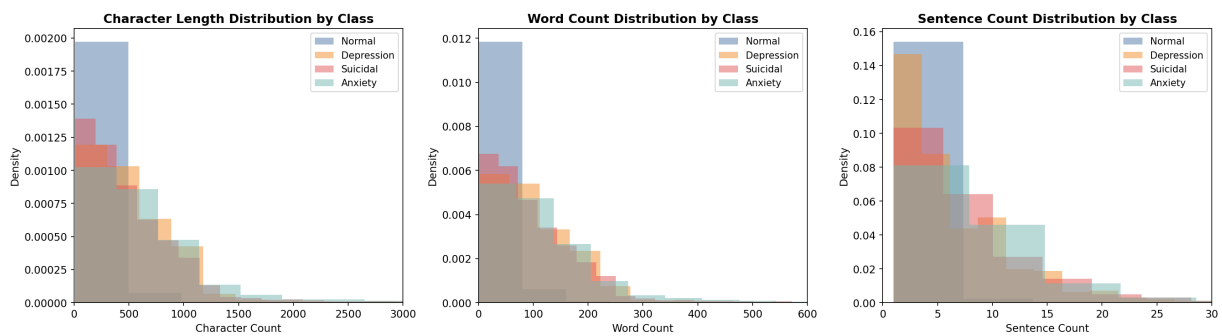


Figure 3-1: Character Length, Word Count, and Sentence Count Distributions by Class

3.2 Text Length Box Plots by Class

The box plot analysis (with outliers removed for visual clarity) confirms the significant length disparity between classes. The interquartile range (IQR) for the Normal class is extremely narrow (6-23 words at Q1-Q3), while mental health categories show much wider IQRs: Depression (47-148 words), Suicidal (38-144 words), and Anxiety (50-157 words). The Anxiety class exhibits the largest median word count (95) and the widest spread, suggesting that anxiety-related posts tend to be the most verbose. This information is critical for setting sequence length parameters in deep learning models: a maximum sequence length of 200-256 tokens would capture approximately 90% of all mental health texts, while a 512-token limit would cover 97-99% of samples across all classes.

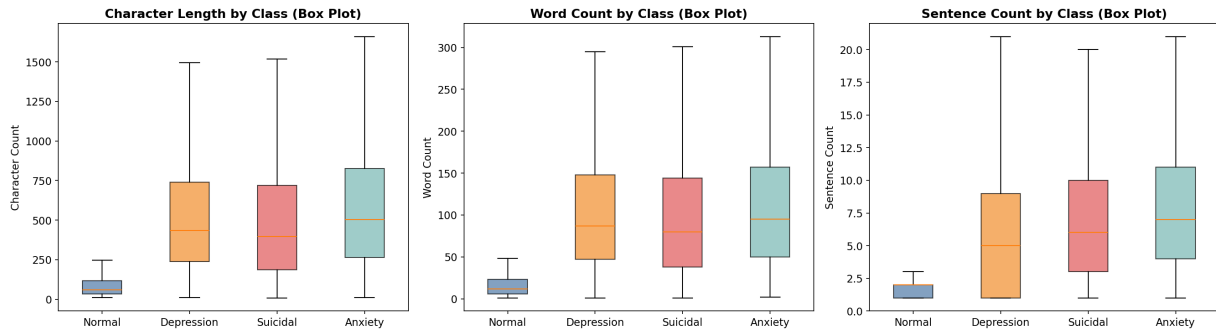


Figure 3-2: Text Length Box Plots by Class (Character, Word, and Sentence Counts)

3.3 Word Count Percentile Analysis

The percentile analysis provides precise guidance for sequence length configuration. At the 90th percentile, mental health texts range from 203 to 223 words, while Normal texts reach only 57 words. At the 99th percentile, Suicidal texts reach 635 words and Anxiety texts reach 688 words, indicating very long outliers. For BERT-based models with a 512-token budget, approximately 1.0-3.2% of mental health texts will exceed the limit, with Anxiety having the highest exceedance rate. For LSTM/GRU models, a maximum sequence length of 300-400 words would balance coverage with computational efficiency. For traditional ML approaches using TF-IDF, the text length variability is less concerning as the vectorization process naturally handles variable-length inputs.

Percentile	Normal	Depression	Suicidal	Anxiety
P10	4	24	19	19
P25	6	47	38	50
P50 (Median)	12	87	80	95
P75	23	148	144	157
P90	57	203	211	223
P95	88	230	251	320
P99	164	409	635	688

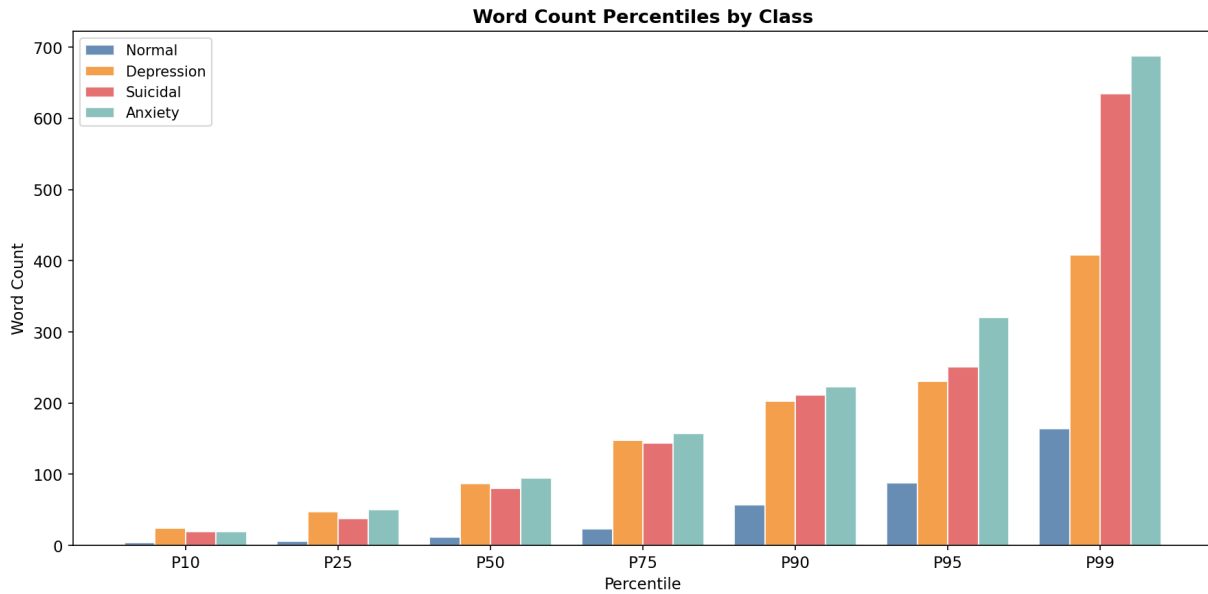


Figure 3-3: Word Count Percentiles by Class

4. Vocabulary & Linguistic Analysis

4.1 Top Words by Class

The top word analysis (after removing English stop words) reveals distinctive vocabulary patterns across classes. The Normal class is dominated by informal and social media language ('mom', 'ur', 'like', 'just'), reflecting its composition of casual social media posts and conversations. The Depression class shows a strong prevalence of emotional and self-referential language ('just', 'like', 'feel', 'life', 'want', 'know', 'time', 'really'), consistent with the introspective and emotionally negative nature of depressive expression. The Suicidal class shares many words with Depression but with notable additions such as 'die', 'kill', and 'want' appearing prominently, indicating more extreme ideation language. The Anxiety class uniquely features 'anxiety' as its most frequent word, along with physical symptom descriptors like 'heart', 'breath', and 'panic', distinguishing it from other categories.

The vocabulary overlap between Depression and Suicidal categories is substantial, sharing high-frequency words like 'just', 'want', 'like', 'life', and 'feel'. This overlap poses a significant challenge for classification models, particularly traditional ML approaches that rely on word-level features. The distinguishing signals between these two classes lie more in intensity markers and specific ideation language (e.g., 'die', 'kill', 'end' for Suicidal versus 'depressed', 'therapy', 'medication' for Depression) rather than in general emotional vocabulary. Transformer-based models like BERT are expected to perform better on this discrimination task due to their ability to capture contextual nuances.

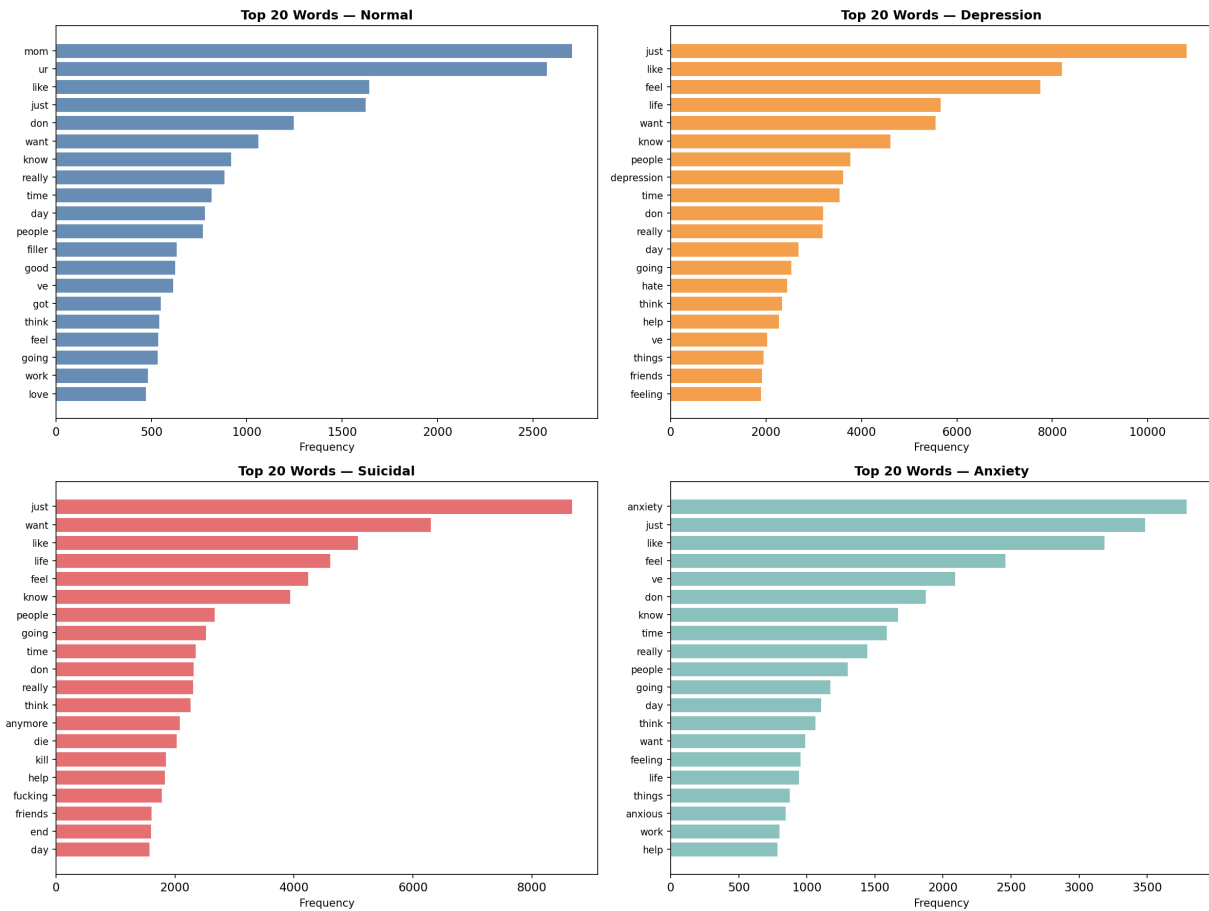


Figure 4-1: Top 20 Words by Class (After Stopword Removal)

4.2 Top Bigrams by Class

Bigram analysis provides more contextual information than unigrams and reveals class-specific phrase patterns. The Normal class features social media-specific bigrams and casual expressions. Depression shows phrases like 'feel like', 'don know', 'want die', and 'mental health', reflecting the typical linguistic patterns of depressive discourse. The Suicidal class exhibits more direct ideation phrases such as 'want die', 'kill myself', and 'end life', which serve as strong discriminative features. Anxiety is characterized by physical symptom bigrams like 'panic attack', 'heart rate', and 'chest pain', alongside anxiety-specific expressions like 'anxiety attack' and 'social anxiety'. These bigram patterns suggest that n-gram features (bigrams and trigrams) would significantly improve traditional ML classifier performance beyond unigram-only approaches.

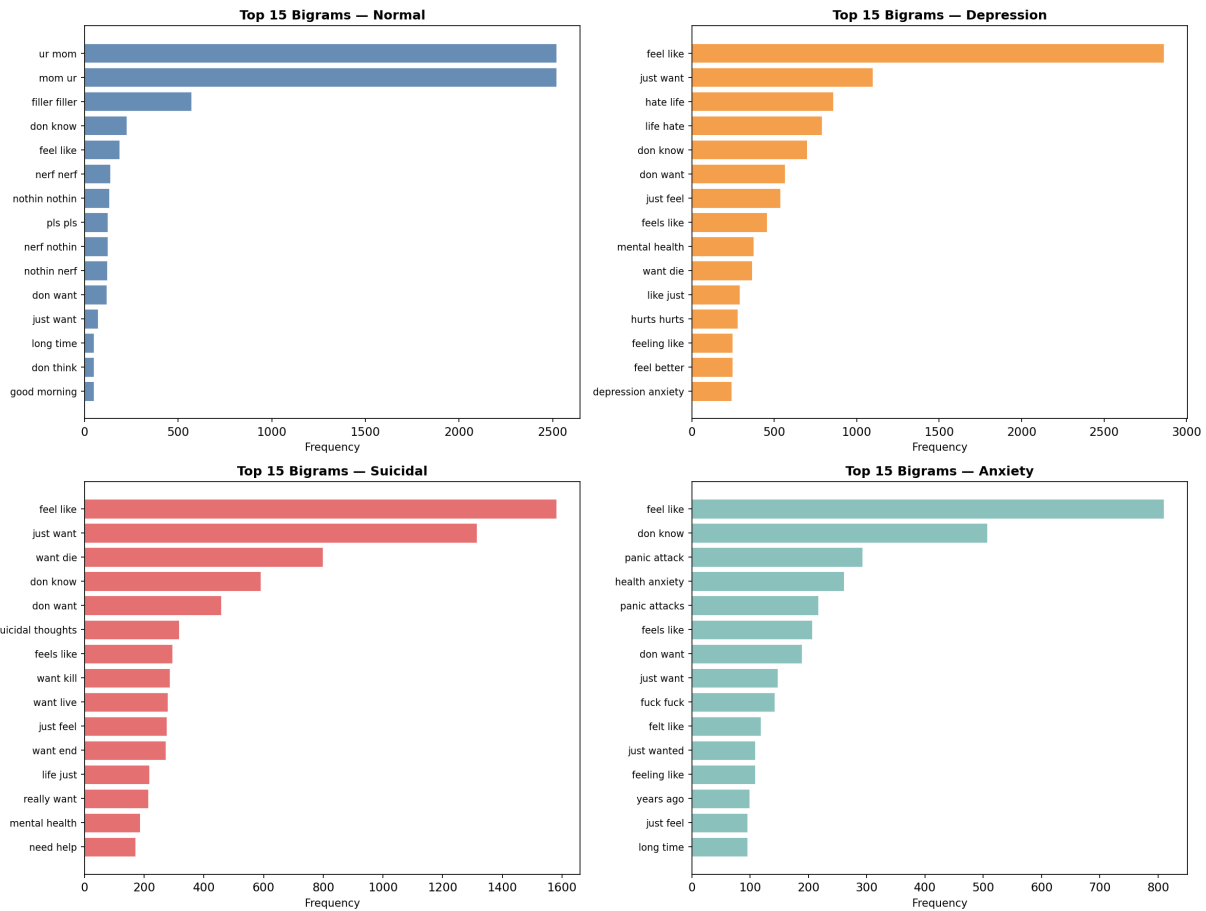


Figure 4-2: Top 15 Bigrams by Class

4.3 TF-IDF Distinctive Words

While raw frequency analysis shows common words shared across classes, TF-IDF analysis highlights the most discriminative terms for each class by weighting terms that are frequent in one class but rare in others. The Normal class TF-IDF analysis highlights informal social media vocabulary and casual terms. Depression's distinctive TF-IDF terms include clinically relevant language and emotional descriptors that are less common in other classes. Suicidal ideation shows the strongest discriminative TF-IDF signals, with terms directly related to self-harm and death ideation receiving high scores. Anxiety's distinctive terms center around physiological symptoms and anxiety-specific terminology. These TF-IDF profiles confirm that while vocabulary overlap exists, each class possesses unique discriminative features that effective models should leverage.

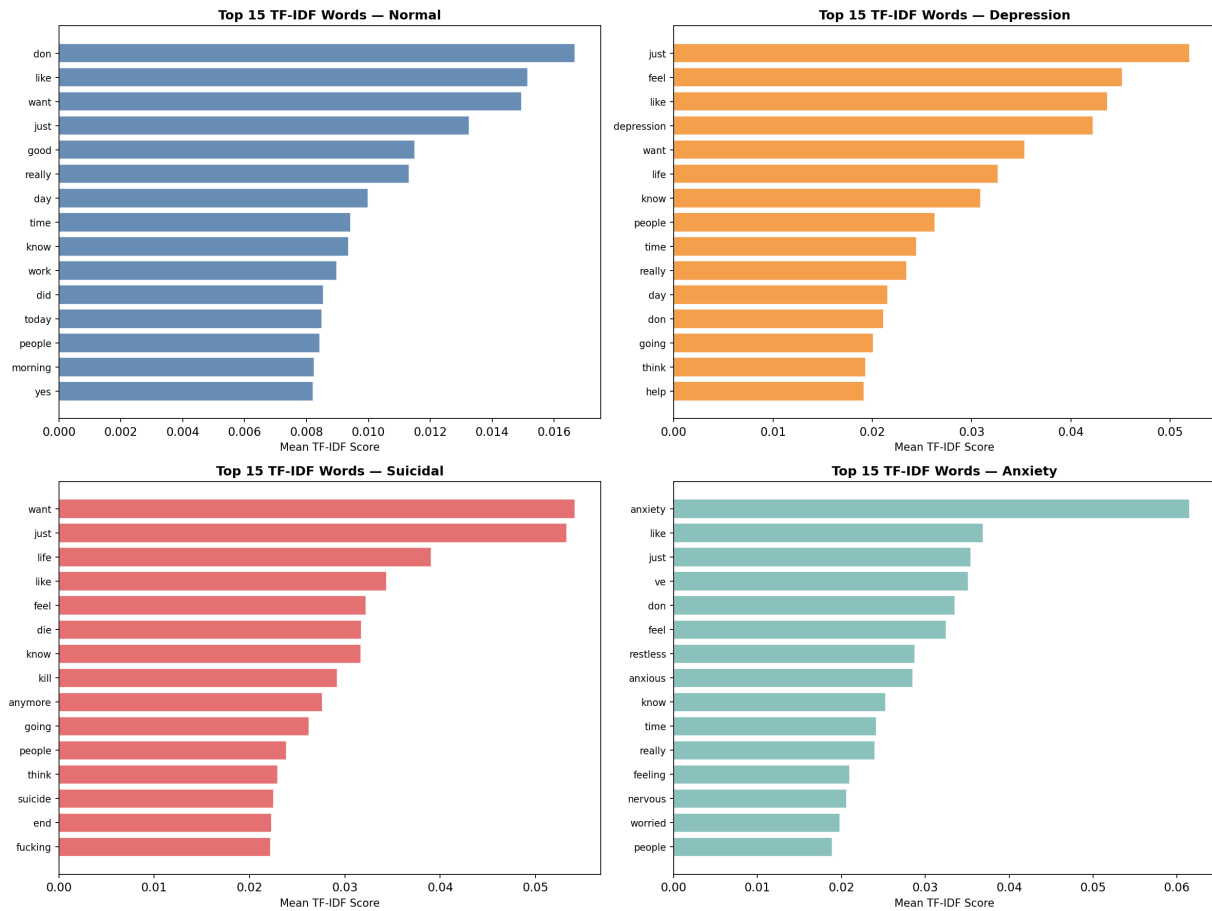


Figure 4-3: Top 15 TF-IDF Distinctive Words by Class

4.4 Vocabulary Overlap Analysis

The vocabulary overlap analysis quantifies the shared vocabulary between classes using both set-based overlap percentages and TF-IDF centroid cosine similarity. The set-based overlap shows that 70-80% of each class's vocabulary is shared with other classes, reflecting the common English vocabulary base. However, the TF-IDF centroid cosine similarity provides a more meaningful measure of semantic similarity. The most critical finding is the extremely high cosine similarity between Depression and Suicidal classes (0.919), indicating that these two categories are nearly indistinguishable at the TF-IDF centroid level. In contrast, Anxiety shows relatively lower similarity with other classes (0.70-0.80), suggesting it is more linguistically distinct. The Normal class has moderate similarity with all mental health classes (0.72-0.77), which is expected given the shared English vocabulary base.

This overlap pattern has significant implications for model selection and evaluation. Binary classifiers between Depression and Suicidal will face the greatest challenge, and model evaluation metrics should particularly track the confusion between these two classes. For multi-class models, the confusion matrix should be carefully analyzed for Depression-Suicidal misclassifications. The relatively distinct Anxiety vocabulary suggests that this class may be easier to identify, but its small sample size (11.3%) creates a different challenge related to class imbalance.

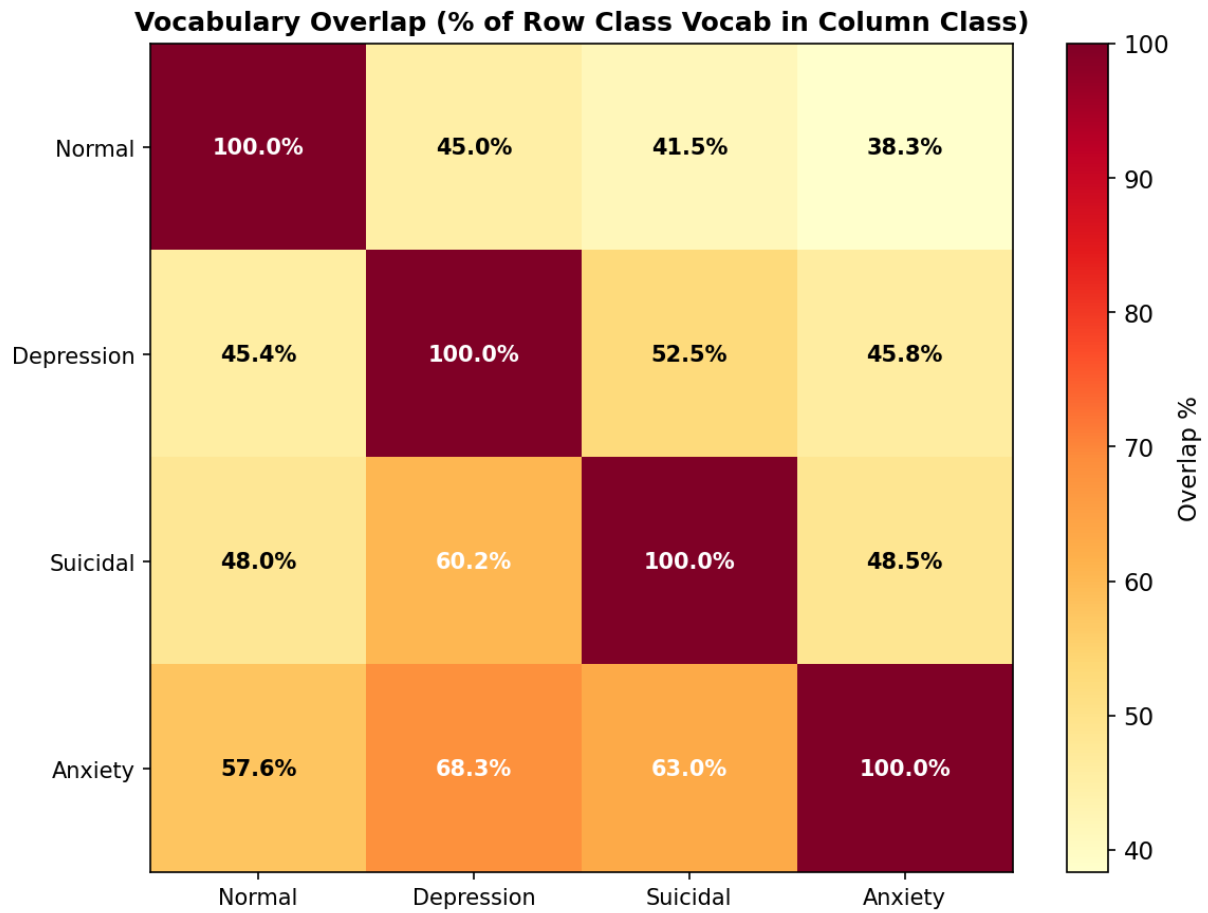


Figure 4-4: Vocabulary Overlap Percentage Between Classes

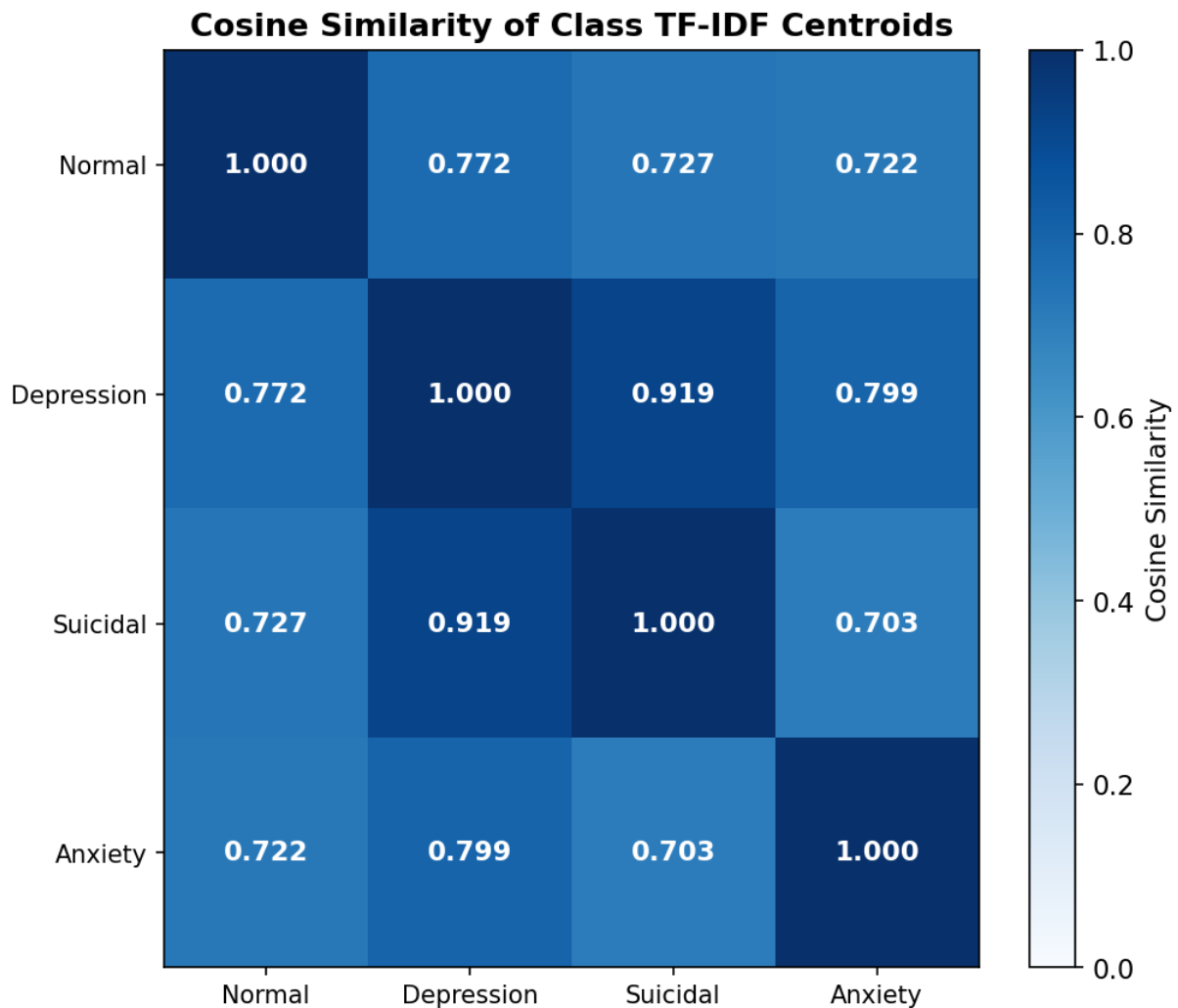


Figure 4-5: Cosine Similarity of TF-IDF Class Centroids

4.5 Lexical Diversity and Word Statistics

Lexical diversity (the ratio of unique words to total words) provides insight into the linguistic richness of each class. Mental health categories tend to have lower lexical diversity compared to the Normal class, likely because emotional distress language involves more repetitive use of certain emotional and self-referential terms (e.g., 'feel', 'just', 'want'). The Normal class, with its diverse range of topics from casual conversation to news sharing, naturally exhibits higher vocabulary variety. Average word length is fairly consistent across classes at approximately 4.5-5.0 characters, though mental health texts may show slightly lower average word length due to the frequent use of short emotional words and contractions. Unique word count per text is substantially higher for mental health categories, consistent with their longer text lengths.

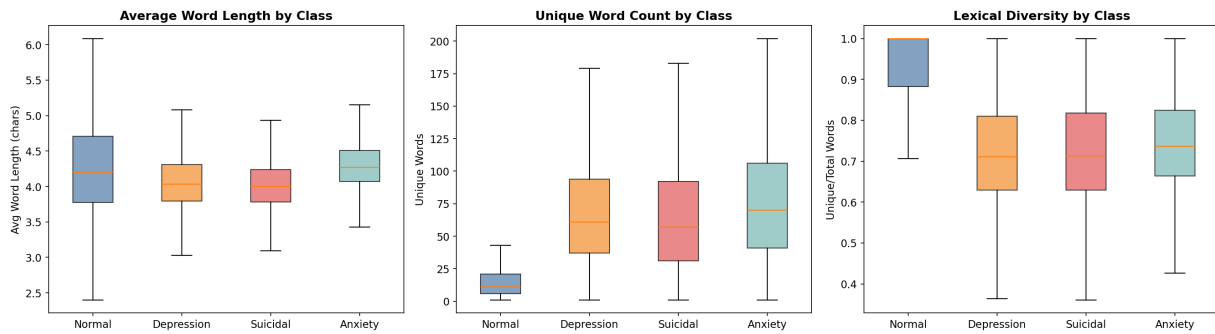


Figure 4-6: Word Statistics by Class (Average Word Length, Unique Words, Lexical Diversity)

5. Feature Correlation Analysis

The feature correlation heatmap reveals important relationships between text-derived features that inform both feature engineering for traditional ML and data preprocessing decisions. As expected, character length and word count are highly correlated ($r > 0.95$), and both are strongly correlated with unique word count, indicating that longer texts naturally contain more unique vocabulary. Lexical diversity shows weak negative correlations with length features, suggesting that longer texts tend to have slightly lower vocabulary diversity ratios, consistent with the tendency for repetitive language in extended emotional expression. Exclamation marks and question marks show moderate positive correlations with each other but weak correlations with length features, suggesting they represent distinct stylistic features independent of text length.

The uppercase ratio shows weak correlations with all other features, indicating it captures a distinct stylistic signal that could be valuable for classification, particularly for distinguishing emotional from neutral text. URL count and ellipsis count show near-zero correlations with most features, confirming they represent independent noise or stylistic signals. For traditional ML feature engineering, the high correlation between character length, word count, and unique word count suggests that including all three would be redundant; selecting one (preferably word count as it is most interpretable) along with less correlated features like exclamation marks, uppercase ratio, and lexical diversity would create a more efficient feature set.

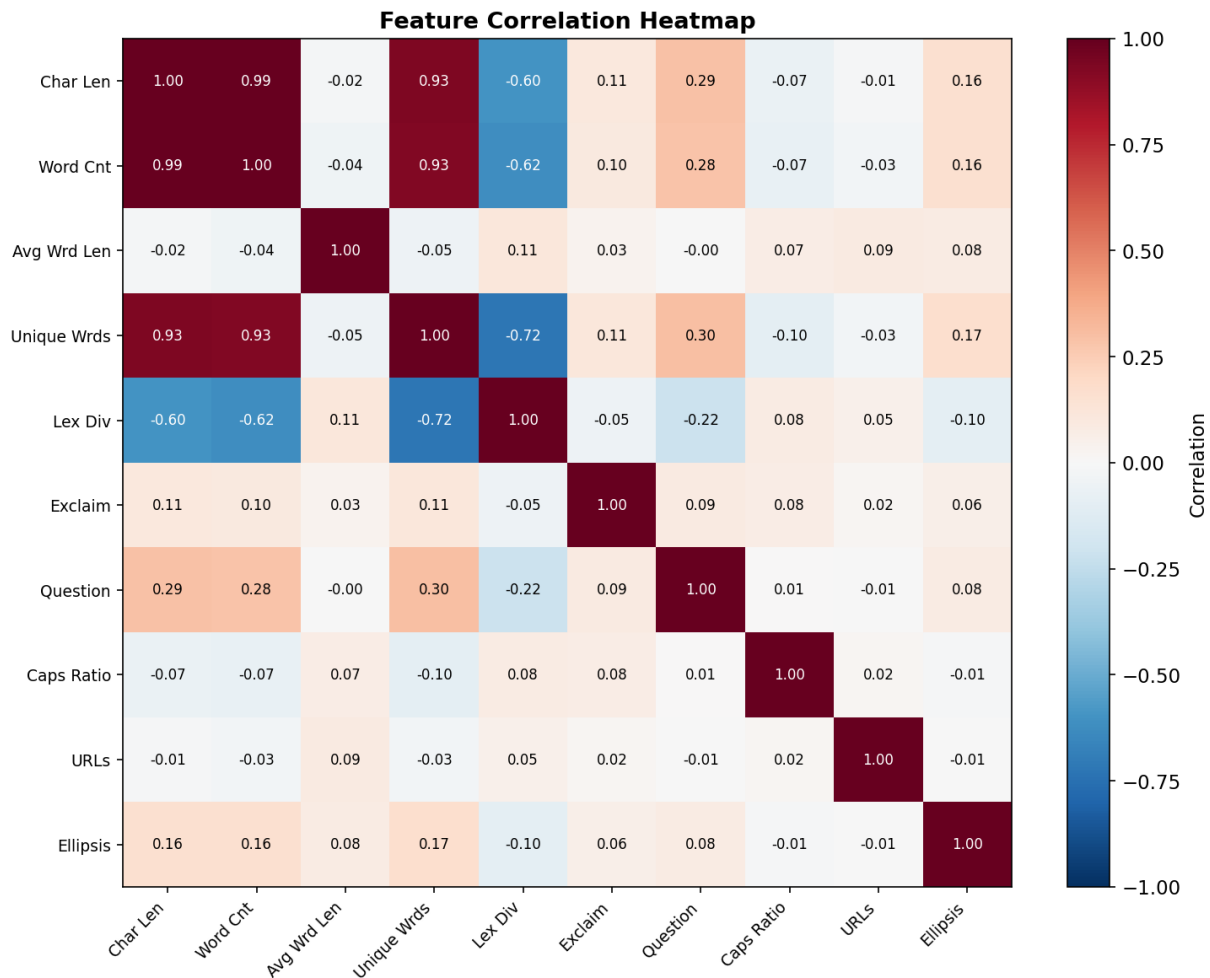


Figure 5-1: Feature Correlation Heatmap

6. Data Quality Issues & Cleaning Recommendations

6.1 Identified Quality Issues

A systematic scan of the dataset reveals several data quality issues that must be addressed before model training. The most prevalent issue is the presence of non-ASCII Unicode characters, affecting 11.48% of samples. These include accented characters, non-Latin scripts, and special symbols that may originate from multilingual social media posts or character encoding artifacts. While some Unicode characters (like emoticons) may carry sentiment information, the majority represent noise for English-focused NLP models. URLs appear in 1.79% of texts, typically from social media links that add no semantic value for mental health classification. Mention tags (@username) affect 1.77% of samples and represent social media artifacts. Repeated characters (5+ consecutive identical characters) appear in 1.70% of texts and often represent emotional emphasis (e.g., 'sooooo', 'ahhhhh') that may carry signal but requires normalization. Very short texts (3 or fewer words) constitute 2.66% of the dataset and are overwhelmingly from the Normal class (850 out of 920), potentially adding noise rather than signal.

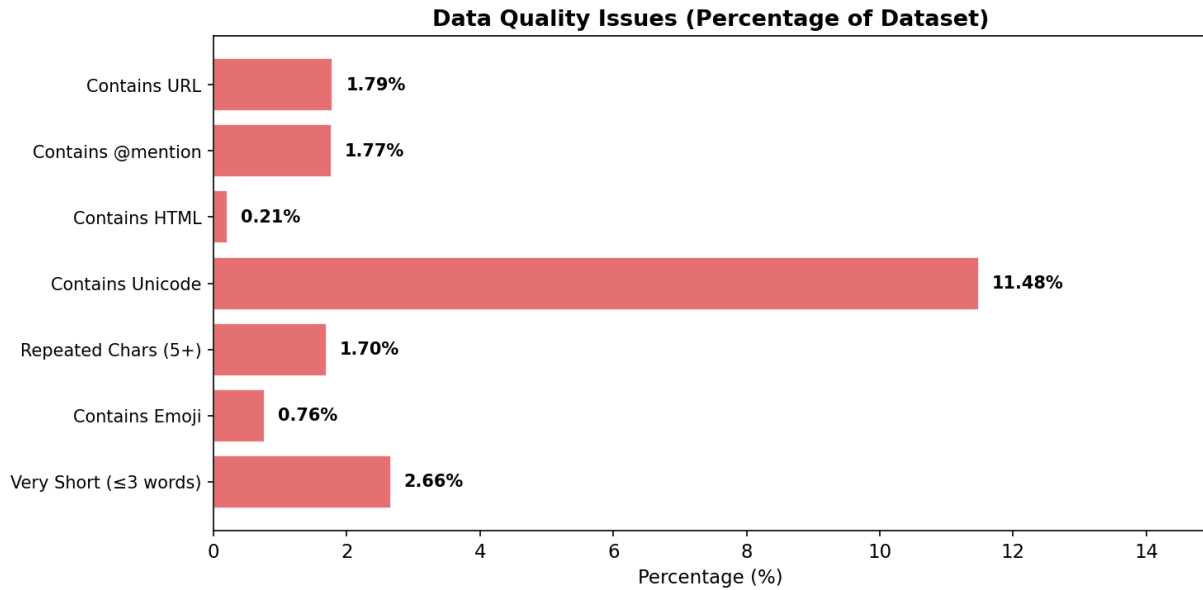


Figure 6-1: Data Quality Issues (Percentage of Dataset)

6.2 Quality Issues by Class

The distribution of quality issues varies significantly across classes. The Normal class has the highest proportion of very short texts (6.5%) and mentions (@username, 3.5%), consistent with its composition of social media posts and tweets. The Anxiety class shows the highest rate of Unicode characters (14.2%) and repeated characters (2.8%), potentially reflecting the more intense emotional expression in anxiety-related posts. URLs are most prevalent in the Normal class (2.5%), suggesting social media link sharing behavior. HTML tags are rare across all classes (0.2%) but should still be handled during preprocessing. The class-specific quality issue patterns suggest that preprocessing should be applied uniformly rather than selectively, as the same issues appear across all classes but at different frequencies.

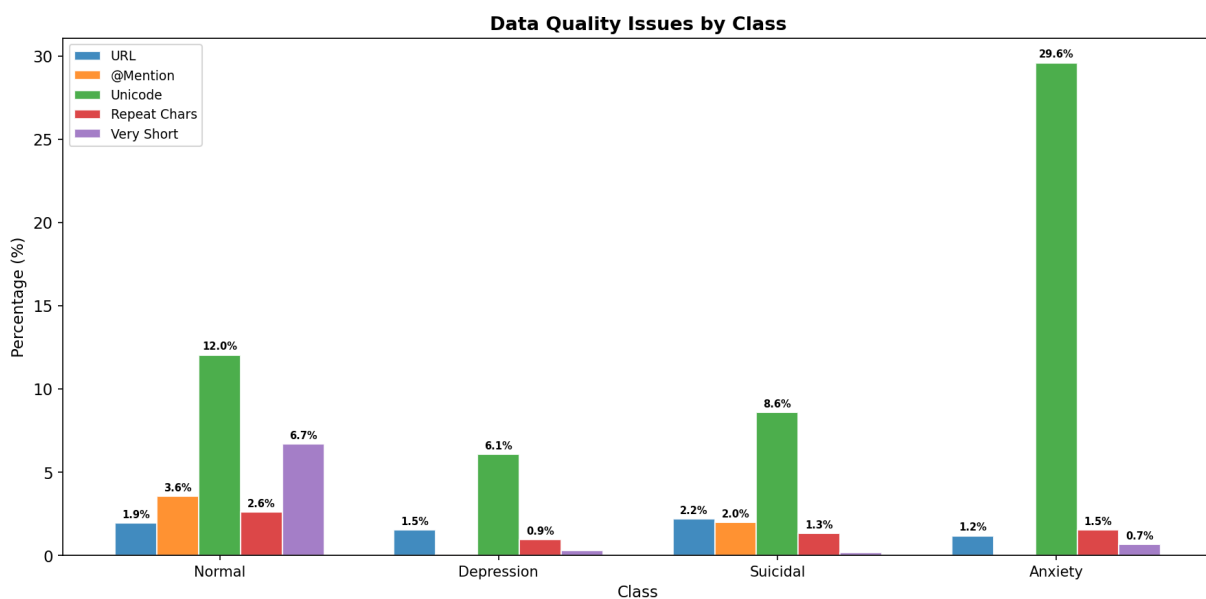


Figure 6-2: Data Quality Issues by Class

6.3 Special Character Analysis

Special character usage patterns provide additional insights into the stylistic differences between classes. Exclamation marks are most prevalent in the Anxiety and Normal classes, reflecting either emotional intensity (Anxiety) or casual social media tone (Normal). Question marks are notably more frequent in Anxiety texts, consistent with the uncertainty and worry that characterize anxiety expression. The uppercase ratio is slightly elevated in Anxiety and Normal classes, potentially indicating emphatic expression. URLs and mentions are predominantly Normal-class features, confirming the social media origin of much Normal-class data. Ellipsis usage is more common in Depression and Suicidal texts, potentially reflecting the trailing off and uncertainty characteristic of depressive discourse. These stylistic signals could complement semantic features in classification models.

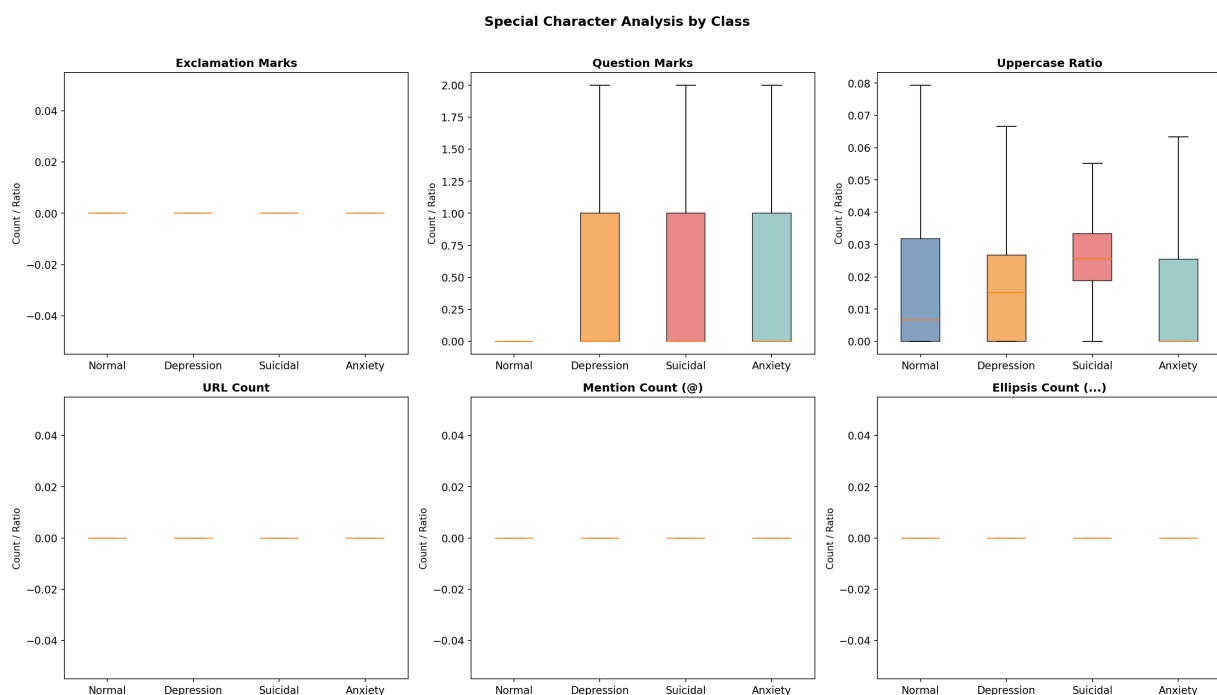


Figure 6-3: Special Character Analysis by Class

6.4 Recommended Data Cleaning Pipeline

Based on the identified quality issues, the following cleaning pipeline is recommended for all three modeling approaches, with approach-specific modifications noted in subsequent sections:

- Remove URLs (`http/https/www` patterns) — replace with space to preserve word boundaries
- Remove @mentions and #hashtags — these are social media artifacts with minimal semantic value
- Remove or normalize HTML entities and tags — decode entities like `&` to `&` where meaningful, strip tags
- Normalize repeated characters (3+ identical consecutive to 2) — e.g., `'sooooo'` to `'soo'`, preserving emphasis signal
- Handle Unicode: normalize accented characters to ASCII equivalents, remove non-Latin scripts

- Strip leading/trailing whitespace and normalize internal whitespace to single spaces
- Consider removing or flagging very short texts (3 or fewer words) — 92% are Normal class and add minimal signal
- Preserve case information for BERT (case-sensitive), but lowercase for traditional ML and some DL models

7. Modeling Recommendations: Traditional Machine Learning

7.1 Feature Engineering Strategy

For traditional ML approaches (Logistic Regression, SVM, Random Forest), the text must be transformed into fixed-length feature vectors. The primary feature extraction method should be TF-IDF vectorization with unigram and bigram features (`ngram_range=(1,2)`), as the bigram analysis in Section 4.2 demonstrated that phrase-level features (e.g., 'panic attack', 'kill myself', 'feel like') carry strong discriminative signals. Sublinear TF scaling (`sublinear_tf=True`) is recommended to dampen the effect of very frequent terms within documents, and minimum document frequency filtering (`min_df=3`) should be applied to remove rare terms that add noise. A maximum feature limit of 50,000-100,000 features balances vocabulary coverage with computational efficiency and overfitting risk.

Beyond TF-IDF, several handcrafted features should be included based on the analysis: (1) Text length features (word count, character count) which show strong class separation; (2) Sentiment lexicon scores using VADER or TextBlob to capture emotional polarity; (3) Special character counts (exclamation marks, question marks, ellipsis) which vary across classes; (4) Uppercase ratio as a stylistic indicator; (5) Lexical diversity as a measure of vocabulary richness; and (6) Mental health keyword counts using curated lexicons for depression, suicidal ideation, and anxiety. These features complement TF-IDF by capturing stylistic and structural signals that bag-of-words approaches may miss.

7.2 Model Selection & Configuration

Three model families are recommended for evaluation: (1) Logistic Regression with L2 regularization (`C=1.0`) as a strong linear baseline, expected to perform well given the high-dimensional sparse TF-IDF features; (2) Linear SVM (SVM with linear kernel) which often outperforms LR on text classification tasks and handles high-dimensional spaces efficiently; and (3) Random Forest or Gradient Boosted Trees (XGBoost/LightGBM) which can capture non-linear feature interactions but may be slower on high-dimensional TF-IDF data. All models should use `class_weight='balanced'` to address the 3.26:1 imbalance ratio. Cross-validation should use stratified k-fold (`k=5`) to ensure representative class distribution in each fold. The primary evaluation metric should be macro F1-score rather than accuracy, as it gives equal weight to all classes regardless of their frequency.

7.3 Expected Challenges

The primary challenge for traditional ML is the high semantic overlap between Depression and Suicidal classes (cosine similarity 0.919). TF-IDF features alone may struggle to distinguish between these categories, as their vocabulary and phrasing overlap significantly. Additional discriminative features such as specific ideation keywords, intensity modifiers, and first-person pronoun usage patterns could help. The moderate class imbalance (Anxiety at 11.3%) may lead to poor minority class recall without proper balancing techniques. The right-skewed text length distribution means that longer texts may dominate the TF-IDF feature space, potentially biasing the model; applying length-based normalization or separate feature scaling is advisable.

8. Modeling Recommendations: Deep Learning

8.1 Architecture Design

For deep learning approaches, an LSTM or BiLSTM architecture is recommended as the primary model, with a CNN-based alternative for comparison. The BiLSTM architecture should consist of: (1) An embedding layer with 300-dimensional embeddings, initialized with pre-trained GloVe or FastText vectors and fine-tuned during training; (2) A bidirectional LSTM layer with 128-256 hidden units per direction, capturing both forward and backward temporal dependencies; (3) Optional attention mechanism on top of the BiLSTM outputs to focus on the most informative parts of the text; (4) A dropout layer (rate 0.3-0.5) for regularization; and (5) A dense output layer with softmax activation for 4-class classification. The attention mechanism is particularly recommended because mental health texts often contain key diagnostic phrases (e.g., 'kill myself', 'panic attack') embedded within longer narratives, and attention can help the model focus on these critical segments.

For the CNN alternative, a multi-kernel architecture (kernel sizes 3, 4, 5) with 128 filters each can capture local n-gram patterns effectively. This architecture may perform particularly well on shorter texts where local phrase patterns are more informative than long-range dependencies. An ensemble of BiLSTM and CNN models could leverage the strengths of both architectures. For both architectures, the maximum sequence length should be set to 256-300 words, which covers approximately 90-95% of all texts across classes, balancing coverage with training efficiency and GPU memory usage.

8.2 Tokenization & Preprocessing

Deep learning models require consistent tokenization and padding. The recommended preprocessing pipeline includes: (1) Apply the standard cleaning pipeline from Section 6.4; (2) Tokenize using the NLTK word tokenizer or spaCy, which handles contractions and punctuation better than simple whitespace splitting; (3) Build a vocabulary from the training set with a minimum frequency threshold of 2, plus special tokens for padding (PAD), unknown words (UNK), and optionally beginning/end of sequence; (4) Pad sequences to the maximum length (256-300 words) with post-padding; and (5) Split the data into train/validation/test sets (80/10/10) with stratified sampling to maintain class proportions. Pre-trained embeddings (GloVe 300d or FastText 300d) should be loaded and used to initialize the embedding layer, with OOV words initialized randomly from a uniform distribution.

8.3 Training Configuration

Key training hyperparameters include: batch size of 32-64 (depending on GPU memory), learning rate of 1e-3 with ReduceLROnPlateau scheduling, Adam optimizer, and categorical cross-entropy loss with class weights inversely proportional to class frequencies (Normal: 0.68, Depression: 0.85, Suicidal: 1.10, Anxiety: 2.21). Early stopping with patience of 5-7 epochs based on validation macro F1-score should prevent overfitting. Gradient clipping (max norm 1.0) is recommended to prevent exploding gradients in LSTM training. Data augmentation techniques such as random synonym replacement, random deletion, and back-translation can help address class imbalance, particularly for the Anxiety class.

9. Modeling Recommendations: Transformer-Based Fine-tuning (BERT)

9.1 Model Selection

For transformer-based fine-tuning, BERT-base-uncased (12 layers, 768 hidden units, 110M parameters) is the recommended starting point. For this mental health classification task, domain-specific considerations suggest that BERT-base should be evaluated alongside mental health-domain models if available. The choice of uncased versus cased model requires careful consideration: while the analysis shows that uppercase ratio varies by class (Section 5), the overall signal is modest, and uncased BERT generally performs well on social media text where case usage is inconsistent. However, a cased model may capture emphasis patterns (e.g., 'I HATE this') that carry emotional signal. If computational resources permit, both should be evaluated.

9.2 Token Length Analysis

The BERT 512-token limit is a critical constraint for this dataset. Our analysis shows that approximately 0.2% of Normal texts, 1.1% of Depression texts, 2.4% of Suicidal texts, and 3.2% of Anxiety texts exceed the approximate 512-token limit (estimated as word count * 1.3). While the overall exceedance rate is low (under 3.2% for any class), the mental health categories have higher exceedance rates than Normal, meaning that truncation will disproportionately affect the classes that are most clinically important. Several strategies can mitigate this: (1) Head-plus-tail truncation (keeping the first 128 and last 382 tokens) which preserves both the opening context and concluding statements; (2) Hierarchical BERT which processes the text in chunks and aggregates representations; or (3) Longformer or BigBird models which support longer sequences (up to 4096 tokens) and could handle the full text length for 99%+ of samples.

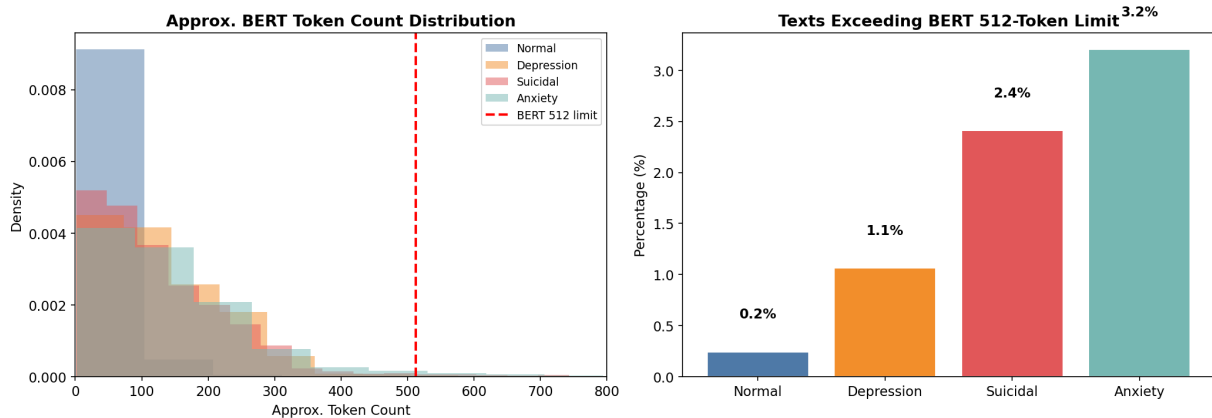


Figure 9-1: Approximate BERT Token Distribution and Exceedance Rates

9.3 Fine-tuning Strategy

The recommended fine-tuning approach follows a two-phase strategy. Phase 1 (feature extraction): Freeze BERT layers and train only the classification head for 3-5 epochs to establish a strong baseline and verify data pipeline correctness. Phase 2 (full fine-tuning): Unfreeze all layers and train with a low learning rate ($2e-5$ to $5e-5$) for 3-4 epochs with early stopping. The learning rate should be applied with a linear warmup for the first 10% of training steps followed by linear decay. Batch size should be 16-32 (limited by GPU memory for BERT), with gradient accumulation if larger effective batch sizes are needed. Class-weighted cross-entropy loss or focal loss should be used to address the 3.26:1 imbalance ratio, with particular attention to the Depression-Suicidal confusion zone identified in Section 4.4.

For evaluation, 5-fold stratified cross-validation is recommended to obtain reliable performance estimates. The primary metric should be macro F1-score, with per-class F1-scores reported to track performance on the minority Anxiety class and the confusable Depression-Suicidal pair. A confusion matrix should be generated for each fold to analyze systematic misclassification patterns. The model achieving the best validation macro F1 should be selected, and its test-set performance reported as the final result. A threshold analysis on the Suicidal class may also be warranted, as the clinical cost of a false negative (missing a suicidal post) is much higher than a false positive.

10. Key Findings & Action Plan

10.1 Critical Findings Summary

The EDA reveals five critical findings that directly impact model design and training decisions. First, the class imbalance of 3.26:1 (Normal to Anxiety) requires explicit handling through class-weighted loss, oversampling, or focal loss across all model architectures. Second, the extremely high TF-IDF centroid similarity between Depression and Suicidal classes (0.919) indicates that these categories are semantically close and will be the primary source of classification errors; models must be specifically designed to capture the subtle linguistic differences between them. Third, text length is a powerful discriminator (Normal median 12 words vs. mental health median 80-95 words), and including length-based features in traditional ML or leveraging the full

context in transformer models will be beneficial. Fourth, approximately 11.5% of texts contain non-ASCII Unicode characters that require normalization, and 2.66% are extremely short texts (mostly Normal class) that may add noise. Fifth, the BERT 512-token limit will truncate 1-3% of mental health texts, with Anxiety having the highest truncation rate, necessitating truncation strategy selection.

10.2 Prioritized Action Plan

Priority	Action	Impact	Risk if Skipped	Validation
P0	Implement class-weighted loss / focal loss	High	Minority class (Anxiety) recall < 50%	Per-class F1 on validation
P0	Handle Depression-Suicidal confusion	High	Critical misclassifications	Confusion matrix analysis
P1	Apply Unicode normalization & URL removal	Medium	Noisy features, degraded performance	Feature quality audit
P1	Set appropriate max sequence lengths	Medium	Information loss or wasted compute	Coverage vs. efficiency
P2	Include bigram features for traditional ML	Medium	Missed phrase-level signals	Unigram vs. bigram ablation
P2	Evaluate attention mechanism for BiLSTM	Medium	Model may miss key phrases	Attention weight visualization
P3	Remove/flag very short texts	Low	Slight noise in Normal class	With vs. without removal
P3	Evaluate head+tail truncation for BERT	Low	2-3% of MH texts truncated	Full vs. head+tail comparison

11. Uncertainty & Limitations

Several limitations should be acknowledged when interpreting this analysis and applying its recommendations. First, the BERT token count analysis uses an approximation (word count * 1.3) rather than actual BERT tokenizer output, which may underestimate or overestimate the true token counts, particularly for texts with many contractions, special characters, or rare words. A precise analysis using the actual BERT tokenizer is recommended before finalizing truncation strategies. Second, the dataset labels may contain annotation noise, particularly for borderline cases between Depression and Suicidal categories where the distinction depends on subjective interpretation of ideation intensity. Inter-annotator agreement scores were not available for this analysis.

Third, the vocabulary and TF-IDF analyses are based on surface-level word matching and do not account for polysemy, sarcasm, or negation, which are prevalent in mental health discourse and may affect model performance in ways not captured by this EDA. Fourth, the dataset appears to be sourced from social media (Reddit, Twitter), which introduces platform-specific biases in language patterns, demographic representation, and expression styles. Models trained on this data may not generalize well to clinical text, therapy transcripts, or other modalities of mental health communication. Fifth, the temporal characteristics of the data are unknown; if the texts span a long time period, concept drift in language usage (especially on social media) may affect model stability.

Future validation steps should include: (1) Running the actual BERT tokenizer on the full dataset to confirm token length distributions; (2) Conducting inter-annotator agreement analysis on a random sample to quantify label noise; (3) Evaluating model performance on an external validation set from a different data source; (4) Implementing fairness analysis across demographic groups if such metadata becomes available; and (5) Establishing a monitoring framework for model performance drift in production deployment scenarios.